

Prepare a project to be deployed

- If your planning to deploy your project to **AWS EKS**, you will find this guide useful. If your planning to deploy your project to any other platform, reach out to any of the [team members](#) and explain your scenario, so they can work on your specific deployment strategy.

There are some prerequisites that have to be met before the DevOps team can start working on the creation of a Jenkins pipeline, make sure that you have met each one of them before you [request the creation of a pipeline](#).

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1. Environment

In order to deploy a project, you need to make sure there's an actual environment to use. If you haven't requested the creation of it, you can do so by opening a ticket of the kind **MANUAL - JENKINS / EKS APPLICATION SETUP**. You will need to open this ticket for dev, qat and production environment. This ticket already includes a task for the pipeline creation at the end, so there's no need to open another ticket to request the pipeline; **you will need to provide the DevOps team with the information of the environment** in the following format.

- If the environment information has not been received after 7 days the pipeline creation task was opened, the task will be closed and marked as completed. You will need to [request the creation of a pipeline](#) in case that happens.

Example

```
1 GITHUB
2 FrontEnd: https://github.avangrid.net/Avangrid-Renewables/FirstTimeRightFrontend.git
3 Backend: https://github.avangrid.net/Avangrid-Renewables/FirstTimeRight.git
4
5 DEV
6 AWS Account name DEV : aws-avangrid-oregon-oprwwdfs-dev
7 AWS Account Number : 637423360040
8 AWS Role: role-av-uswest2-oprwwdfs_eksadmins-dev
9 EKS Cluster: eks-av-uswest2-oprwwdfs-dev
10
11 QAT
12 AWS Account name INT: aws-avangrid-oregon-oprwwdfs-int
13 AWS Account Number : 381491858142
14 AWS Role: role-av-uswest2-oprwwdfs_eksadmins-int
```

```
15 EKS Cluster: eks-av-uswest2-oprwddefs-int
16
17 PROD
18 AWS Account name PR0 : aws-avangrid-oregon-oprwddefs-pro
19 AWS Account Number : 51725332055
20 AWS Role: role-av-uswest2-oprwddefs_eksadmins-pro
21 EKS Cluster: eks-av-uswest2-oprwddefs-pro
22
23 ECR
24 Dev & QAT: 443472960778.dkr.ecr.us-west-2.amazonaws.com/ab-ftright/tright
25 Prod: 545390015186.dkr.ecr.us-west-2.amazonaws.com/ab-first-time-right/ftirwcoas
26
27 KUBERNETES NAMESPACE
28 ab-ftright-intranet
```

2. Branching strategy

Your project must contain the branches necessary for the pipeline to work, Jenkins will look for these specific branches in order to deploy to the different environments. You can have N number of branches in your repository - you can include feature and releases branches - just make sure you include one of the necessary branches for each deployment environment.

- **Development:** development, develop or dev.
- **QAT:** release or qat.
- **Production:** production, main or master.

 Just include one for each deployment environment

3. Functional code

You can request the creation of a pipeline if there's no code in the repository. If there's no functional version of the app yet, what you can do is push a dummy app to the repository.

4. Specific requirements for each technology

Depending on the technology your project uses (C#, Java, Python, Nodejs, etc), your repository will need to contain specific files in it. Go to the page that corresponds to the technology you are using to learn more about these files.

Checklist for a new pipeline - pre-requisites that DevOps team needs before we can start working on the new pipeline

1. Pre-requisites before request to create pipeline is open – these items should be completed –

Application Team.

- Environments
- Branching Strategy
- Code/application

- d. Repositories
- e. Target Platform (Aws, Azure, etc)
- f. Architecture approval (for any app that depends on new or non-compliant/standard technologies)

2. **Application Team** needs to provide

- a. Yaml file
- b. Target deploy domain
- c. Container registry
- d. Compilation instructions or Dockerfile

3. Request to create pipeline can be created by **Application Team**. NOTE: DevOps may need up to 7 business days to create a standard pipeline. If pipeline is not standard – unique technology or specific tools that need to be use – timeframe needs to be reassessed on case-by-case basis

4. At the same time **Application Team** should contact **Cloud** team for

- a. Secrets
- b. Gateway

5. **DevOps** will start working on the pipeline

6. **DevOps** will test pipeline

7. **Application Team** will perform detailed testing. Sign off will be required when testing is completed in each environment.

8. **Application Team** will sign off on the completed work

[Link to the check list](#)

- [Angular](#)
- [Java Springboot](#)
- [Nodejs](#)
- [Python](#)